

SDEV2501

Lesson 22 – Exploratory Analysis

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Today's objective

- Understand the **purpose** of exploratory data analysis (EDA)
- Perform basic exploratory analysis using visualizations

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Agenda

- What is **exploratory data analysis**?
- Getting to know your **data**
- **Univariate**, **bivariate**, and **multivariate** analysis
- Asking better exploratory **questions**
- **Interpreting** and **documenting** observations

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EDA

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Imagine you've just been handed a dataset with 10,000 rows and 15 columns. What's the first thing you do?

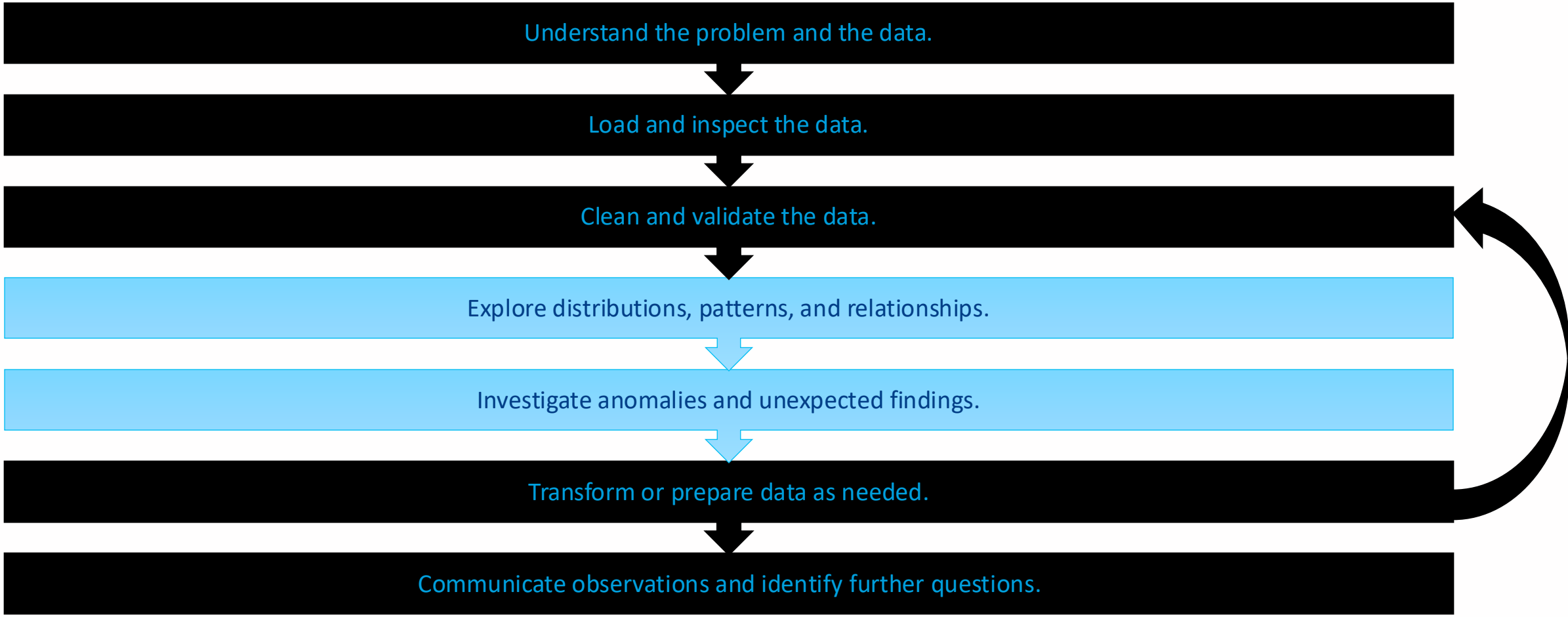
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What is EDA?

- EDA is examining the data **thoroughly** to find its
  - **Characteristics** (e.g. # of features, data types, data distribution, measures of central tendency, measures of variability)
  - Possible **anomalies** (e.g. errors or outliers)
  - Hidden **patterns** and **relationships**
- EDA is **iterative**: one discovery often leads to another question

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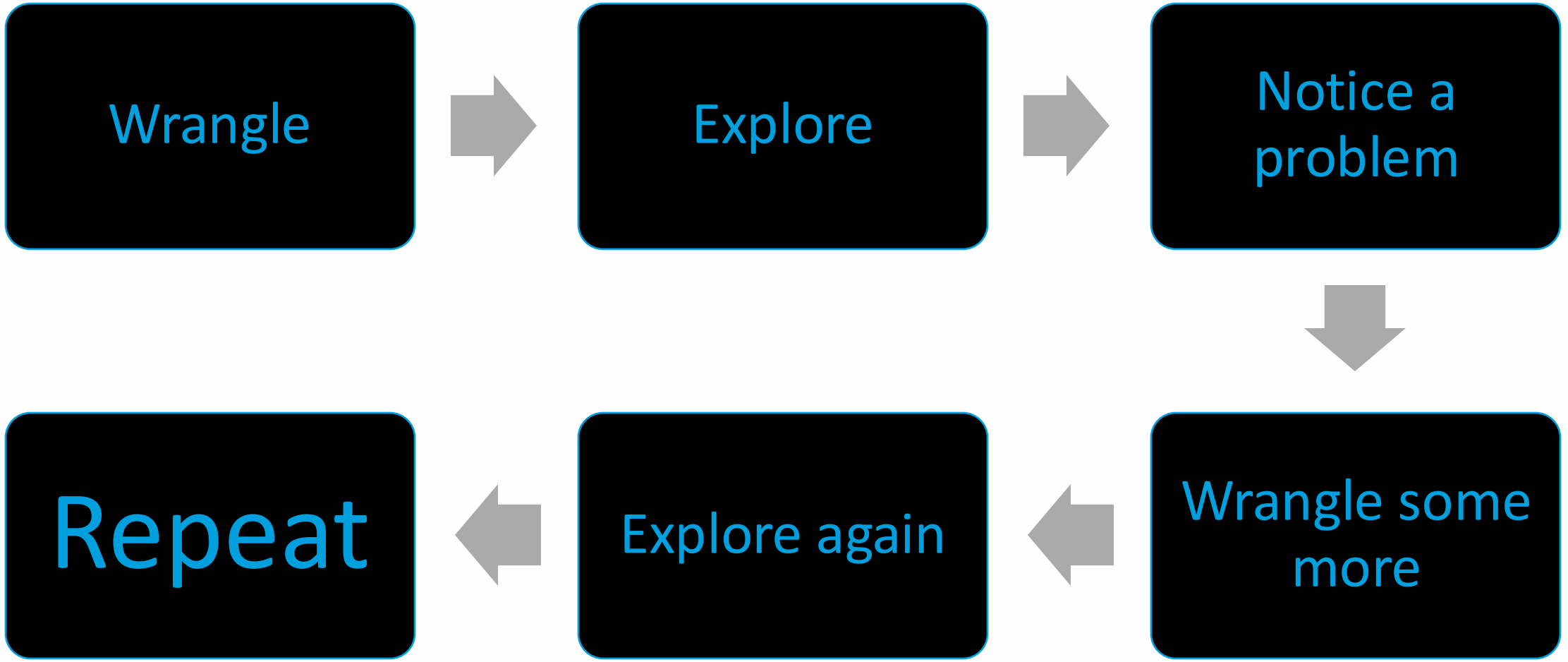
Where does EDA fit in our life cycle?



How is EDA different from data wrangling?

	Data Wrangling	EDA
	“Can I analyze this data?”	“What can I learn from this data?”
Main purpose	**Prepare** and **restructure** data	**Explore** and **understand** data
Focus	The **data** itself	The **patterns** within the data
Typical questions	Is the data in the right format? Do tables need to be joined? Do columns need to be reshaped?	What is typical? What varies? Are there patterns, relationships, clusters, or unusual observations?
Typical activities	Merge/join datasets, reshape, pivot/unpivot, create variables, split/ combine columns, aggregate, filter	Descriptive statistics, distributions, correlations, group comparisons, visualizations, identifying patterns and anomalies
Output	An analysis-ready **dataset**	**Insights** and **questions** about the dataset

Our workflow is NOT linear



Start by getting to know your data:

- ❑How many **rows** and **columns** are there?
- ❑What does each **row** represent?
- ❑What does each **column** represent?
- ❑Which columns are **numeric**, **categorical**, or **dates**?
- ❑Are there **missing** values?
- ❑Are there **duplicate** records?
- ❑Are there **unexpected** or **inconsistent** values?
- ❑Are the values within reasonable **ranges**?

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What should we look for?

Category	What are we investigating?
Distributions	How are values spread out?
Anomalies	Are there unusual or unexpected values?
Patterns	Are there trends, clusters, or recurring behaviours?
Relationships	Do variables appear to be connected?

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Example of EDA iteration

Question

- Are orders larger in certain months?

Explore

- Visualize monthly order quantities

Observe

- December appears to have higher quantities.

New **question**

- Is this true across all categories?

Explore again!

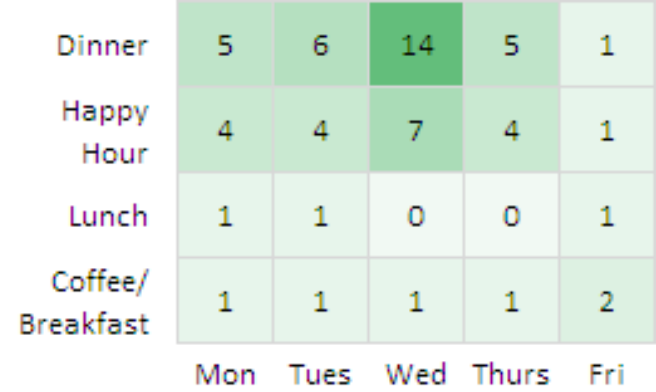
Univariate Analysis

Studying **one variable** at a time to understand its characteristics & distribution.

- **Histograms** show how data values are distributed.
  - “Are values clustered or spread out? Is the distribution skewed?”
- **Box plots** help detect outliers and show data spread.
  - “What's the median? Are there potential outliers?”
- **Bar charts** are used for categorical variables.
  - “Which category is most common? Are some categories underrepresented?”

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Bivariate Analysis



Examines the relationship between **two variables** to understand how they interact or influence each other.

Variable Combination	Example	Visualization
Numeric + Numeric	Quantity vs unit cost	Scatter plot
Numeric + Categorical	Order value by membership type	Box plot
Categorical + Categorical	Membership type by city	Grouped bar chart / crosstab

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Bivariate Analysis, cont'd

- **Correlation** coefficients measure the strength of the relationship.
  - e.g. Pearson correlation
    - Close to +1 means a strong positive linear relationship
    - Close to 0 means little/no relationship
    - Close to -1 means a strong negative linear relationship

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Multivariate Analysis

Studies **3+ variables** together to understand more complex relationships.

For example:

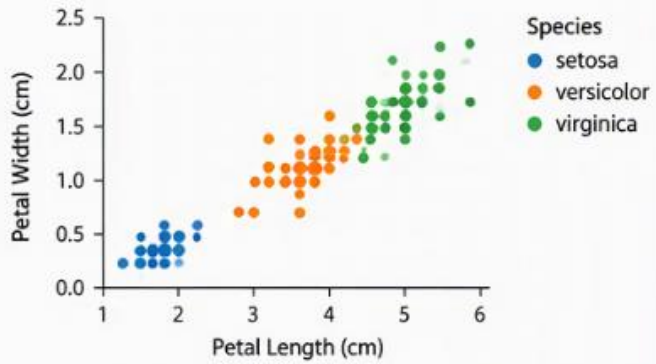
- Scatter plots using **colour** to represent a third variable.
- Bubble charts using **size** to represent another variable.
- **Grouped** visualizations comparing several categories.
- Heatmaps showing **relationships** between multiple numeric variables.
- Pair plots to explore several numeric variables simultaneously.

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Visualizations that use 3+ variables together

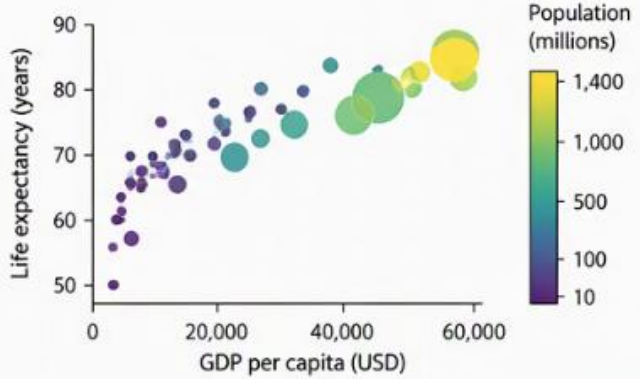
These plots help us see more complex relationships in the data.

1. Scatter plot with colour (3 variables)



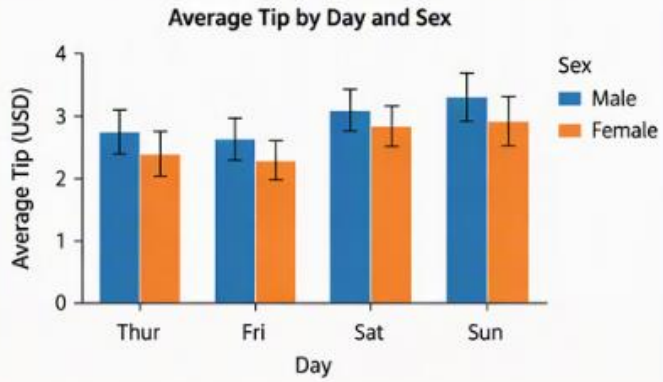
Each point is a flower. The x-axis shows petal length, the y-axis shows petal width, and colour shows species (a third variable).

2. Bubble chart with size (3 variables)



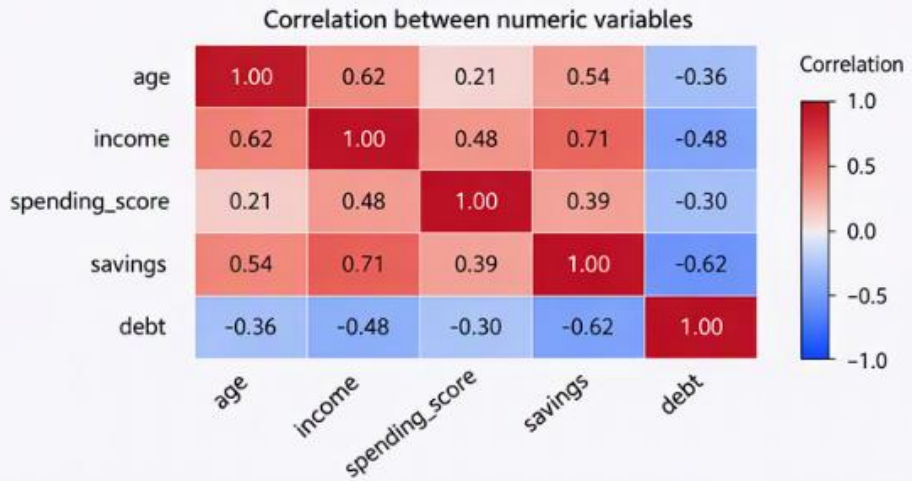
Each point is a country. The x-axis shows GDP per capita, the y-axis shows life expectancy, the size and colour of the bubbles show population (a third variable).

3. Grouped visualization (3 variables)



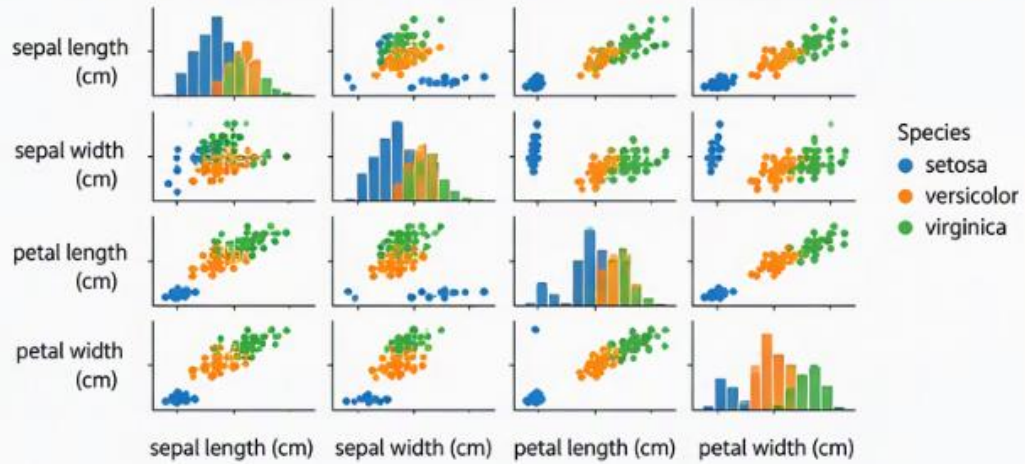
Compares average tip amount (y-axis) across days of the week (x-axis), with a separate bar for each sex (a third variable).

4. Heatmap (relationships between multiple variables)



Shows the correlation (relationship) between several numeric variables at once. Darker red = strong positive relationship, darker blue = strong negative relationship.

5. Pair plot (several numeric variables simultaneously)



Shows scatter plots and distributions for several numeric variables at once. Each point is colour-coded by species (a third variable).

Exercise: What's the appropriate viz?

1. What does the **distribution** of order quantities look like?
2. Which product categories are most **popular**?
3. Are order **quantities** related to **unit cost**?
4. Are order **values** different across **membership t**ypes?
5. How do **monthly orders** change over time?
6. Are there unusually **large** orders?

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EDA helps us identify things worth investigating. It doesn't automatically explain why they happen.

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For example:

A histogram shows that **most customer orders** contain **between 1 and 5 items**, but a small number contain **more than 20**.

We can reasonably say that **most orders are relatively small**, with a **few unusually large** orders.

We **cannot** conclude that large orders are placed by business customers.

What we should investigate next?

- **Who** placed the large orders?
- Are they associated with a particular **membership t**ype?
- Are these **legitimate orders** or **data entry errors**?

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Practice

Given a dataset:

- Identify **three things** you want to know about the dataset.
- Identify two potential **data quality concerns**.
- Write five **questions** you would like to investigate.
- Identify **which columns** might be useful for answering those questions.

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EDA challenge

Given a dataset, complete a mini EDA:

- ❑ Identify the **number** of rows and columns.
- ❑ Identify at least three different **variable types**.
- ❑ Calculate **descriptive statistics** for two numeric variables.
- ❑ Create **two visualizations** using different chart types.
- ❑ Identify at least **two observations**.
- ❑ Formulate **one new question** based on something they discovered.

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Closing

- Write 3 examples of exploratory questions.
- 1-minute reflection: What surprised you during exploration?

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What’s next?

- Explanatory Analysis

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